catalogue_q9_iso0 over GF(9)

June 23, 2026

The equation

The equation of the quartic curve is :

$$X_1^4 + 6X_0^3X_2 + 4X_0X_2^3$$

$$(0, 1, 0, 0, 6, 0, 0, 4, 0, 0, 0, 0, 0, 0, 0)$$

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X1^4 + 6*X0^3*X2 + 4*X0*X2^3
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General information

Number of bitangents	28
Number of points	28
Fullness	is full
Number of Kovalevski points	63
Bitangent line type (a_0, a_1, a_2)	$(0, 28, 0)$
Number of singular points	0
Stabilizer order	12096
Stabilizer order invariant	1, 2^{315}, 3^{728}, 4^{756}, 6^{2520}, 7^{1728}, 8^{3024}, 12^{3024}
Action on points	28
Action on bitangents	28
Action on Kovalevski pts	63

Automorphism group:
Strong generators for a group of order 12096:

$$\left[\begin{array}{ccc} 2 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 1 \end{array}\right]_0, \left[\begin{array}{ccc} 7 & 0 & 0 \\ 0 & 5 & 0 \\ 0 & 0 & 1 \end{array}\right]_1, \left[\begin{array}{ccc} 7 & 0 & 0 \\ 0 & 1 & 0 \\ 6 & 0 & 1 \end{array}\right]_1, \left[\begin{array}{ccc} 7 & 0 & 0 \\ 8 & 2 & 0 \\ 2 & 1 & 1 \end{array}\right]_1, \left[\begin{array}{ccc} 0 & 0 & 1 \\ 0 & 2 & 0 \\ 1 & 0 & 0 \end{array}\right]_1$$

1,0,0,0,6,0,0,0,2,0
1,0,0,0,2,0,0,0,5,1

1,0,0,0,5,0,8,0,5,1
1,0,0,3,7,0,7,5,5,1
0,0,1,0,2,0,1,0,0,1

All points on the curve

The quartic has 28 points:
The points on the quartic curve are:

$0: P_0 = (1,0,0)$
$1: P_2 = (0,0,1)$
$2: P_{15} = (4,0,1)$
$3: P_{19} = (8,0,1)$
$4: P_{21} = (2,1,1)$
$5: P_{22} = (3,1,1)$
$6: P_{26} = (7,1,1)$
$7: P_{30} = (2,2,1)$
$8: P_{31} = (3,2,1)$
$9: P_{35} = (7,2,1)$
$10: P_{38} = (1,3,1)$
$11: P_{42} = (5,3,1)$
$12: P_{43} = (6,3,1)$
$13: P_{47} = (1,4,1)$
$14: P_{51} = (5,4,1)$
$15: P_{52} = (6,4,1)$
$16: P_{57} = (2,5,1)$
$17: P_{58} = (3,5,1)$
$18: P_{62} = (7,5,1)$
$19: P_{65} = (1,6,1)$
$20: P_{69} = (5,6,1)$
$21: P_{70} = (6,6,1)$
$22: P_{75} = (2,7,1)$
$23: P_{76} = (3,7,1)$
$24: P_{80} = (7,7,1)$
$25: P_{83} = (1,8,1)$
$26: P_{87} = (5,8,1)$
$27: P_{88} = (6,8,1)$

The points by rank are: (0, 2, 15, 19, 21, 22, 26, 30, 31, 35, 38, 42, 43, 47, 51, 52, 57, 58, 62, 65, 69, 70, 75, 76, 80, 83, 87, 88)
The Kovalevski points are:
$0: P_3 = (1,1,1) = a_1 \cap a_3 \cap c_{16} \cap c_{36}$
$1: P_4 = (1,1,0) = a_2 \cap b_5 \cap c_{25} \cap d$
$2: P_5 = (2,1,0) = a_6 \cap b_1 \cap c_{16} \cap d$
$3: P_6 = (3,1,0) = a_5 \cap b_3 \cap c_{35} \cap d$
$4: P_7 = (4,1,0) = a_1 \cap b_4 \cap c_{14} \cap d$
$5: P_8 = (5,1,0) = c_{13} \cap c_{26} \cap c_{45} \cap d$
$6: P_9 = (6,1,0) = a_4 \cap b_6 \cap c_{46} \cap d$
$7: P_{10} = (7,1,0) = c_{15} \cap c_{24} \cap c_{36} \cap d$
$8: P_{11} = (8,1,0) = a_3 \cap b_2 \cap c_{23} \cap d$
$9: P_{12} = (1,0,1) = a_5 \cap b_6 \cap c_{14} \cap c_{23}$
$10: P_{13} = (2,0,1) = c_{15} \cap c_{16} \cap c_{25} \cap c_{26}$
$11: P_{14} = (3,0,1) = a_2 \cap b_1 \cap c_{36} \cap c_{45}$
$12: P_{16} = (5,0,1) = a_3 \cap a_4 \cap b_3 \cap b_4$
$13: P_{17} = (6,0,1) = a_1 \cap b_2 \cap c_{35} \cap c_{46}$
$14: P_{18} = (7,0,1) = a_6 \cap b_5 \cap c_{13} \cap c_{24}$
$15: P_{20} = (0,1,1) = c_{12} \cap c_{13} \cap c_{25} \cap c_{35}$
$16: P_{23} = (4,1,1) = a_2 \cap a_5 \cap c_{26} \cap c_{56}$
$17: P_{24} = (5,1,1) = b_1 \cap b_2 \cap c_{14} \cap c_{24}$
$18: P_{25} = (6,1,1) = a_6 \cap b_4 \cap c_{15} \cap c_{23}$
$19: P_{27} = (8,1,1) = b_3 \cap b_5 \cap c_{34} \cap c_{45}$
$20: P_{28} = (0,2,1) = c_{12} \cap c_{16} \cap c_{24} \cap c_{46}$

$21: P_{29} = (1,2,1) = b_2 \cap b_4 \cap c_{25} \cap c_{45}$
$22: P_{32} = (4,2,1) = b_1 \cap b_6 \cap c_{15} \cap c_{56}$
$23: P_{33} = (5,2,1) = a_1 \cap a_2 \cap c_{13} \cap c_{23}$
$24: P_{34} = (6,2,1) = a_3 \cap b_5 \cap c_{14} \cap c_{26}$
$25: P_{36} = (8,2,1) = a_4 \cap a_6 \cap c_{34} \cap c_{36}$
$26: P_{37} = (0,3,1) = b_1 \cap b_3 \cap c_{12} \cap c_{23}$
$27: P_{39} = (2,3,1) = c_{13} \cap c_{14} \cap c_{36} \cap c_{46}$
$28: P_{40} = (3,3,1) = b_4 \cap b_6 \cap c_{24} \cap c_{26}$
$29: P_{41} = (4,3,1) = a_3 \cap a_6 \cap c_{35} \cap c_{56}$
$30: P_{44} = (7,3,1) = a_1 \cap a_4 \cap c_{15} \cap c_{45}$
$31: P_{45} = (8,3,1) = a_5 \cap b_2 \cap c_{16} \cap c_{34}$
$32: P_{46} = (0,4,1) = a_3 \cap b_6 \cap c_{12} \cap c_{45}$
$33: P_{48} = (2,4,1) = a_1 \cap a_5 \cap b_1 \cap b_5$
$34: P_{49} = (3,4,1) = a_6 \cap b_3 \cap c_{14} \cap c_{25}$
$35: P_{50} = (4,4,1) = a_4 \cap b_2 \cap c_{13} \cap c_{56}$
$36: P_{53} = (7,4,1) = a_2 \cap b_4 \cap c_{16} \cap c_{35}$
$37: P_{54} = (8,4,1) = c_{23} \cap c_{26} \cap c_{34} \cap c_{46}$
$38: P_{55} = (0,5,1) = b_2 \cap b_5 \cap c_{12} \cap c_{15}$
$39: P_{56} = (1,5,1) = a_4 \cap b_1 \cap c_{26} \cap c_{35}$
$40: P_{59} = (4,5,1) = c_{23} \cap c_{25} \cap c_{36} \cap c_{56}$
$41: P_{60} = (5,5,1) = a_5 \cap a_6 \cap c_{45} \cap c_{46}$
$42: P_{61} = (6,5,1) = b_3 \cap b_6 \cap c_{13} \cap c_{16}$
$43: P_{63} = (8,5,1) = a_2 \cap a_3 \cap c_{24} \cap c_{34}$
$44: P_{64} = (0,6,1) = a_2 \cap a_4 \cap c_{12} \cap c_{14}$
$45: P_{66} = (2,6,1) = c_{23} \cap c_{24} \cap c_{35} \cap c_{45}$
$46: P_{67} = (3,6,1) = a_3 \cap a_5 \cap c_{13} \cap c_{15}$
$47: P_{68} = (4,6,1) = b_4 \cap b_5 \cap c_{46} \cap c_{56}$
$48: P_{71} = (7,6,1) = b_2 \cap b_3 \cap c_{26} \cap c_{36}$
$49: P_{72} = (8,6,1) = a_1 \cap b_6 \cap c_{25} \cap c_{34}$
$50: P_{73} = (0,7,1) = a_1 \cap a_6 \cap c_{12} \cap c_{26}$
$51: P_{74} = (1,7,1) = a_2 \cap b_3 \cap c_{15} \cap c_{46}$
$52: P_{77} = (4,7,1) = c_{14} \cap c_{16} \cap c_{45} \cap c_{56}$
$53: P_{78} = (5,7,1) = b_5 \cap b_6 \cap c_{35} \cap c_{36}$
$54: P_{79} = (6,7,1) = a_4 \cap a_5 \cap c_{24} \cap c_{25}$
$55: P_{81} = (8,7,1) = b_1 \cap b_4 \cap c_{13} \cap c_{34}$
$56: P_{82} = (0,8,1) = a_5 \cap b_4 \cap c_{12} \cap c_{36}$
$57: P_{84} = (2,8,1) = a_2 \cap a_6 \cap b_2 \cap b_6$
$58: P_{85} = (3,8,1) = a_4 \cap b_5 \cap c_{16} \cap c_{23}$
$59: P_{86} = (4,8,1) = a_1 \cap b_3 \cap c_{24} \cap c_{56}$
$60: P_{89} = (7,8,1) = a_3 \cap b_1 \cap c_{25} \cap c_{46}$
$61: P_{90} = (8,8,1) = c_{14} \cap c_{15} \cap c_{34} \cap c_{35}$
$62: P_1 = (0,1,0) = c_{12} \cap c_{34} \cap c_{56} \cap d$
The incidence structure induced by the Kovalevski points is:
Flags=(0, 4, 13, 23, 30, 33, 49, 50, 59, 64, 74, 79, 86, 99, 106, 107, 114, 120, 126, 134, 138, 150, 155, 158, 169, 172, 186, 195, 201, 214, 219, 224, 228, 233, 243, 247, 255, 261, 268, 283, 285, 293, 298, 306, 308, 317, 329, 333, 340, 344,

349, 356, 365, 372, 380, 389, 395, 400, 404, 411, 417, 433, 438, 449, 454, 458, 462, 472, 476, 479, 489, 498, 507, 516, 523, 530, 538, 546, 552, 555, 563, 571, 579, 585, 588, 595, 603, 614, 622, 623, 631, 644, 649, 654, 663, 668, 677, 683, 688, 699, 702, 715, 721, 725, 735, 742, 746, 750, 771, 776, 782, 788, 794, 800, 806, 812, 818, 824, 833, 834, 842, 846, 854, 861, 865, 874, 886, 891, 899, 906, 909, 916, 926, 934, 943, 952, 955, 963, 967, 975, 983, 991, 996, 1006, 1008, 1010, 1018, 1028, 1039, 1044, 1050, 1060, 1066, 1079, 1080, 1089, 1094, 1097, 1108, 1111, 1116, 1129, 1141, 1148, 1151, 1154, 1162, 1177, 1179, 1188, 1193, 1198, 1207, 1212, 1218, 1231, 1237, 1246, 1251, 1257, 1265, 1270, 1276, 1284, 1288, 1297, 1299, 1308, 1310, 1342, 1348, 1354, 1360, 1366, 1372, 1378, 1384, 1385, 1389, 1399, 1401, 1415, 1422, 1425, 1431, 1439, 1447, 1449, 1456, 1460, 1474, 1476, 1489, 1497, 1502, 1505, 1517, 1523, 1531, 1533, 1542, 1544, 1553, 1557, 1564, 1581, 1588, 1595, 1602, 1612, 1616, 1622, 1626, 1635, 1654, 1660, 1667, 1673, 1678, 1685, 1690, 1697, 1700, 1702, 1703, 1704, 1705, 1706, 1707, 1708, 1709, 1763)

nb_rows=28

nb_cols=63

nb_flags=252

Incma=

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(0, 0, 1, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 1, 0, 0, 0, 1, 0, 0, 0, 0, 0, 0, 1, 0, 0, 0, 1, 0, 0, 0, 0, 1, 0, 0, 0, 0, 0, 0, 1, 0, 0, 0, 0, 0, 0, 0, 0, 1, 0, 0, 0, 0, 0, 0, 1, 0, 0, 0, 0, 0)

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)
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)

The Kovalevski points sorted by rank are: (1, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 16, 17, 18, 20, 23, 24, 25, 27, 28, 29, 32, 33, 34, 36, 37, 39, 40, 41, 44, 45, 46, 48, 49, 50, 53, 54, 55, 56, 59, 60, 61, 63, 64, 66, 67, 68, 71, 72, 73, 74, 77, 78, 79, 81, 82, 84, 85, 86, 89, 90)

The points off the curve are:

$0: P_1 = (0,1,0)$
$1: P_3 = (1,1,1)$
$2: P_4 = (1,1,0)$
$3: P_5 = (2,1,0)$
$4: P_6 = (3,1,0)$
$5: P_7 = (4,1,0)$
$6: P_8 = (5,1,0)$
$7: P_9 = (6,1,0)$
$8: P_{10} = (7,1,0)$
$9: P_{11} = (8,1,0)$
$10: P_{12} = (1,0,1)$
$11: P_{13} = (2,0,1)$
$12: P_{14} = (3,0,1)$
$13: P_{16} = (5,0,1)$
$14: P_{17} = (6,0,1)$
$15: P_{18} = (7,0,1)$
$16: P_{20} = (0,1,1)$
$17: P_{23} = (4,1,1)$
$18: P_{24} = (5,1,1)$
$19: P_{25} = (6,1,1)$
$20: P_{27} = (8,1,1)$
$21: P_{28} = (0,2,1)$
$22: P_{29} = (1,2,1)$
$23: P_{32} = (4,2,1)$
$24: P_{33} = (5,2,1)$
$25: P_{34} = (6,2,1)$
$26: P_{36} = (8,2,1)$
$27: P_{37} = (0,3,1)$
$28: P_{39} = (2,3,1)$
$29: P_{40} = (3,3,1)$
$30: P_{41} = (4,3,1)$
$31: P_{44} = (7,3,1)$
$32: P_{45} = (8,3,1)$
$33: P_{46} = (0,4,1)$
$34: P_{48} = (2,4,1)$
$35: P_{49} = (3,4,1)$
$36: P_{50} = (4,4,1)$
$37: P_{53} = (7,4,1)$
$38: P_{54} = (8,4,1)$
$39: P_{55} = (0,5,1)$
$40: P_{56} = (1,5,1)$
$41: P_{59} = (4,5,1)$
$42: P_{60} = (5,5,1)$
$43: P_{61} = (6,5,1)$
$44: P_{63} = (8,5,1)$
$45: P_{64} = (0,6,1)$
$46: P_{66} = (2,6,1)$
$47: P_{67} = (3,6,1)$
$48: P_{68} = (4,6,1)$
$49: P_{71} = (7,6,1)$
$50: P_{72} = (8,6,1)$
$51: P_{73} = (0,7,1)$
$52: P_{74} = (1,7,1)$
$53: P_{77} = (4,7,1)$
$54: P_{78} = (5,7,1)$
$55: P_{79} = (6,7,1)$
$56: P_{81} = (8,7,1)$
$57: P_{82} = (0,8,1)$
$58: P_{84} = (2,8,1)$
$59: P_{85} = (3,8,1)$
$60: P_{86} = (4,8,1)$
$61: P_{89} = (7,8,1)$
$62: P_{90} = (8,8,1)$

(1, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 16, 17, 18, 20, 23, 24, 25, 27, 28, 29, 32, 33, 34, 36, 37, 39, 40, 41, 44, 45, 46, 48, 49, 50, 53, 54, 55, 56, 59, 60, 61, 63, 64, 66, 67, 68, 71, 72, 73, 74, 77, 78, 79, 81, 82, 84, 85, 86, 89, 90)

Duals of Bitangents

dual of bitangents:
$0: P_{77} = (4, 7, 1)$
$1: P_{54} = (8, 4, 1)$
$2: P_{69} = (5, 6, 1)$
$3: P_{51} = (5, 4, 1)$
$4: P_{39} = (2, 3, 1)$
$5: P_{80} = (7, 7, 1)$
$6: P_{90} = (8, 8, 1)$
$7: P_{59} = (4, 5, 1)$
$8: P_{87} = (5, 8, 1)$
$9: P_{42} = (5, 3, 1)$
$10: P_{62} = (7, 5, 1)$
$11: P_{66} = (2, 6, 1)$
$12: P_{0} = (1, 0, 0)$
$13: P_{35} = (7, 2, 1)$
$14: P_{48} = (2, 4, 1)$
$15: P_{56} = (1, 5, 1)$
$16: P_{3} = (1, 1, 1)$
$17: P_{84} = (2, 8, 1)$
$18: P_{26} = (7, 1, 1)$
$19: P_{29} = (1, 2, 1)$
$20: P_{74} = (1, 7, 1)$
$21: P_{14} = (3, 0, 1)$
$22: P_{32} = (4, 2, 1)$
$23: P_{45} = (8, 3, 1)$
$24: P_{72} = (8, 6, 1)$
$25: P_{23} = (4, 1, 1)$
$26: P_{17} = (6, 0, 1)$
$27: P_{2} = (0, 0, 1)$
Dimension of the vanishing ideal is 1

```
8*X1^4 + 5*X0^3*X2 + 2*X0*X2^3
```

The lines and their points of contact are:

$$a_1 = \begin{bmatrix} 1 & 0 & 8 \\ 0 & 1 & 5 \end{bmatrix}_{85} = \begin{bmatrix} 1 & 0 & 8 \\ 0 & 1 & 5 \end{bmatrix}_{85} P_{58} = \mathbf{P}(3, 5, 1)\ 4\times$$
$$a_2 = \begin{bmatrix} 1 & 0 & 4 \\ 0 & 1 & 8 \end{bmatrix}_{48} = \begin{bmatrix} 1 & 0 & 4 \\ 0 & 1 & 8 \end{bmatrix}_{48} P_{43} = \mathbf{P}(6, 3, 1)\ 4\times$$
$$a_3 = \begin{bmatrix} 1 & 0 & 7 \\ 0 & 1 & 3 \end{bmatrix}_{73} = \begin{bmatrix} 1 & 0 & 7 \\ 0 & 1 & 3 \end{bmatrix}_{73} P_{75} = \mathbf{P}(2, 7, 1)\ 4\times$$
$$a_4 = \begin{bmatrix} 1 & 0 & 7 \\ 0 & 1 & 8 \end{bmatrix}_{78} = \begin{bmatrix} 1 & 0 & 7 \\ 0 & 1 & 8 \end{bmatrix}_{78} P_{21} = \mathbf{P}(2, 1, 1)\ 4\times$$
$$a_5 = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 6 \end{bmatrix}_{16} = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 6 \end{bmatrix}_{16} P_{35} = \mathbf{P}(7, 2, 1)\ 4\times$$

$$a_6 = \begin{bmatrix} 1 & 0 & 5 \\ 0 & 1 & 5 \end{bmatrix}_{55} = \begin{bmatrix} 1 & 0 & 5 \\ 0 & 1 & 5 \end{bmatrix}_{55} P_{65} = \mathbf{P}(1,6,1)\ 4\times$$
$$b_1 = \begin{bmatrix} 1 & 0 & 4 \\ 0 & 1 & 4 \end{bmatrix}_{44} = \begin{bmatrix} 1 & 0 & 4 \\ 0 & 1 & 4 \end{bmatrix}_{44} P_{70} = \mathbf{P}(6,6,1)\ 4\times$$
$$b_2 = \begin{bmatrix} 1 & 0 & 8 \\ 0 & 1 & 7 \end{bmatrix}_{87} = \begin{bmatrix} 1 & 0 & 8 \\ 0 & 1 & 7 \end{bmatrix}_{87} P_{76} = \mathbf{P}(3,7,1)\ 4\times$$
$$b_3 = \begin{bmatrix} 1 & 0 & 7 \\ 0 & 1 & 4 \end{bmatrix}_{74} = \begin{bmatrix} 1 & 0 & 7 \\ 0 & 1 & 4 \end{bmatrix}_{74} P_{30} = \mathbf{P}(2,2,1)\ 4\times$$
$$b_4 = \begin{bmatrix} 1 & 0 & 7 \\ 0 & 1 & 6 \end{bmatrix}_{76} = \begin{bmatrix} 1 & 0 & 7 \\ 0 & 1 & 6 \end{bmatrix}_{76} P_{57} = \mathbf{P}(2,5,1)\ 4\times$$
$$b_5 = \begin{bmatrix} 1 & 0 & 5 \\ 0 & 1 & 7 \end{bmatrix}_{57} = \begin{bmatrix} 1 & 0 & 5 \\ 0 & 1 & 7 \end{bmatrix}_{57} P_{38} = \mathbf{P}(1,3,1)\ 4\times$$
$$b_6 = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 3 \end{bmatrix}_{13} = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 3 \end{bmatrix}_{13} P_{26} = \mathbf{P}(7,1,1)\ 4\times$$
$$c_{12} = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}_{90} = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}_{90} P_{2} = \mathbf{P}(0,0,1)\ 4\times$$
$$c_{13} = \begin{bmatrix} 1 & 0 & 5 \\ 0 & 1 & 1 \end{bmatrix}_{51} = \begin{bmatrix} 1 & 0 & 5 \\ 0 & 1 & 1 \end{bmatrix}_{51} P_{83} = \mathbf{P}(1,8,1)\ 4\times$$
$$c_{14} = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 8 \end{bmatrix}_{18} = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 8 \end{bmatrix}_{18} P_{62} = \mathbf{P}(7,5,1)\ 4\times$$
$$c_{15} = \begin{bmatrix} 1 & 0 & 2 \\ 0 & 1 & 7 \end{bmatrix}_{27} = \begin{bmatrix} 1 & 0 & 2 \\ 0 & 1 & 7 \end{bmatrix}_{27} P_{51} = \mathbf{P}(5,4,1)\ 4\times$$
$$c_{16} = \begin{bmatrix} 1 & 0 & 2 \\ 0 & 1 & 2 \end{bmatrix}_{22} = \begin{bmatrix} 1 & 0 & 2 \\ 0 & 1 & 2 \end{bmatrix}_{22} P_{69} = \mathbf{P}(5,6,1)\ 4\times$$
$$c_{23} = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 4 \end{bmatrix}_{14} = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 4 \end{bmatrix}_{14} P_{80} = \mathbf{P}(7,7,1)\ 4\times$$
$$c_{24} = \begin{bmatrix} 1 & 0 & 5 \\ 0 & 1 & 2 \end{bmatrix}_{52} = \begin{bmatrix} 1 & 0 & 5 \\ 0 & 1 & 2 \end{bmatrix}_{52} P_{47} = \mathbf{P}(1,4,1)\ 4\times$$
$$c_{25} = \begin{bmatrix} 1 & 0 & 2 \\ 0 & 1 & 1 \end{bmatrix}_{21} = \begin{bmatrix} 1 & 0 & 2 \\ 0 & 1 & 1 \end{bmatrix}_{21} P_{42} = \mathbf{P}(5,3,1)\ 4\times$$
$$c_{26} = \begin{bmatrix} 1 & 0 & 2 \\ 0 & 1 & 5 \end{bmatrix}_{25} = \begin{bmatrix} 1 & 0 & 2 \\ 0 & 1 & 5 \end{bmatrix}_{25} P_{87} = \mathbf{P}(5,8,1)\ 4\times$$
$$c_{34} = \begin{bmatrix} 1 & 0 & 6 \\ 0 & 1 & 0 \end{bmatrix}_{60} = \begin{bmatrix} 1 & 0 & 6 \\ 0 & 1 & 0 \end{bmatrix}_{60} P_{19} = \mathbf{P}(8,0,1)\ 4\times$$
$$c_{35} = \begin{bmatrix} 1 & 0 & 8 \\ 0 & 1 & 1 \end{bmatrix}_{81} = \begin{bmatrix} 1 & 0 & 8 \\ 0 & 1 & 1 \end{bmatrix}_{81} P_{31} = \mathbf{P}(3,2,1)\ 4\times$$
$$c_{36} = \begin{bmatrix} 1 & 0 & 4 \\ 0 & 1 & 6 \end{bmatrix}_{46} = \begin{bmatrix} 1 & 0 & 4 \\ 0 & 1 & 6 \end{bmatrix}_{46} P_{52} = \mathbf{P}(6,4,1)\ 4\times$$
$$c_{45} = \begin{bmatrix} 1 & 0 & 4 \\ 0 & 1 & 3 \end{bmatrix}_{43} = \begin{bmatrix} 1 & 0 & 4 \\ 0 & 1 & 3 \end{bmatrix}_{43} P_{88} = \mathbf{P}(6,8,1)\ 4\times$$
$$c_{46} = \begin{bmatrix} 1 & 0 & 8 \\ 0 & 1 & 2 \end{bmatrix}_{82} = \begin{bmatrix} 1 & 0 & 8 \\ 0 & 1 & 2 \end{bmatrix}_{82} P_{22} = \mathbf{P}(3,1,1)\ 4\times$$

$c_{56} = \left[\begin{array}{ccc} 1 & 0 & 3 \\ 0 & 1 & 0 \end{array}\right]_{30} = \left[\begin{array}{ccc} 1 & 0 & 3 \\ 0 & 1 & 0 \end{array}\right]_{30} P_{15} = \mathbf{P}(4,0,1)\ 4\times$
$d = \left[\begin{array}{ccc} 1 & 0 & 0 \\ 0 & 1 & 0 \end{array}\right]_{0} = \left[\begin{array}{ccc} 1 & 0 & 0 \\ 0 & 1 & 0 \end{array}\right]_{0} P_{0} = \mathbf{P}(1,0,0)\ 4\times$
Rank of lines: (85, 48, 73, 78, 16, 55, 44, 87, 74, 76, 57, 13, 90, 51, 18, 27, 22, 14, 52, 21, 25, 60, 81, 46, 43, 82, 30, 0)
Line type: 1^{28}

28	1	$1,2,3,4,5,6,7,8,9,10,$ $11,12,13,14,15,16,17,18,19,20,$ $21,22,23,24,25,26,27,0$

point types: 1^{28}

28	1	$1,2,3,4,5,6,7,8,9,10,$ $11,12,13,14,15,16,17,18,19,20,$ $21,22,23,24,25,26,27,0$

point types for points off the curve: 4^{63}

63	4	$1,2,3,4,5,6,7,8,9,10,$ $11,12,13,14,15,16,17,18,19,20,$ $21,22,23,24,25,26,27,28,29,30,$ $31,32,33,34,35,36,37,38,39,40,$ $41,42,43,44,45,46,47,48,49,50,$ $51,52,53,54,55,56,57,58,59,60,$ $61,62,0$

Lines on points off the curve:
Off point 0 $= P_1 = (0,1,0)$ lies on 4 bisecants : { 12, 21, 26, 27 }
Off point 1 $= P_3 = (1,1,1)$ lies on 4 bisecants : { 0, 2, 16, 23 }
Off point 2 $= P_4 = (1,1,0)$ lies on 4 bisecants : { 1, 10, 19, 27 }
Off point 3 $= P_5 = (2,1,0)$ lies on 4 bisecants : { 5, 6, 16, 27 }
Off point 4 $= P_6 = (3,1,0)$ lies on 4 bisecants : { 4, 8, 22, 27 }
Off point 5 $= P_7 = (4,1,0)$ lies on 4 bisecants : { 0, 9, 14, 27 }
Off point 6 $= P_8 = (5,1,0)$ lies on 4 bisecants : { 13, 20, 24, 27 }
Off point 7 $= P_9 = (6,1,0)$ lies on 4 bisecants : { 3, 11, 25, 27 }
Off point 8 $= P_{10} = (7,1,0)$ lies on 4 bisecants : { 15, 18, 23, 27 }
Off point 9 $= P_{11} = (8,1,0)$ lies on 4 bisecants : { 2, 7, 17, 27 }
Off point 10 $= P_{12} = (1,0,1)$ lies on 4 bisecants : { 4, 11, 14, 17 }
Off point 11 $= P_{13} = (2,0,1)$ lies on 4 bisecants : { 15, 16, 19, 20 }
Off point 12 $= P_{14} = (3,0,1)$ lies on 4 bisecants : { 1, 6, 23, 24 }
Off point 13 $= P_{16} = (5,0,1)$ lies on 4 bisecants : { 2, 3, 8, 9 }
Off point 14 $= P_{17} = (6,0,1)$ lies on 4 bisecants : { 0, 7, 22, 25 }
Off point 15 $= P_{18} = (7,0,1)$ lies on 4 bisecants : { 5, 10, 13, 18 }
Off point 16 $= P_{20} = (0,1,1)$ lies on 4 bisecants : { 12, 13, 19, 22 }
Off point 17 $= P_{23} = (4,1,1)$ lies on 4 bisecants : { 1, 4, 20, 26 }
Off point 18 $= P_{24} = (5,1,1)$ lies on 4 bisecants : { 6, 7, 14, 18 }

Off point 19 = $P_{25} = (6, 1, 1)$ lies on 4 bisecants : { 5, 9, 15, 17 }
Off point 20 = $P_{27} = (8, 1, 1)$ lies on 4 bisecants : { 8, 10, 21, 24 }
Off point 21 = $P_{28} = (0, 2, 1)$ lies on 4 bisecants : { 12, 16, 18, 25 }
Off point 22 = $P_{29} = (1, 2, 1)$ lies on 4 bisecants : { 7, 9, 19, 24 }
Off point 23 = $P_{32} = (4, 2, 1)$ lies on 4 bisecants : { 6, 11, 15, 26 }
Off point 24 = $P_{33} = (5, 2, 1)$ lies on 4 bisecants : { 0, 1, 13, 17 }
Off point 25 = $P_{34} = (6, 2, 1)$ lies on 4 bisecants : { 2, 10, 14, 20 }
Off point 26 = $P_{36} = (8, 2, 1)$ lies on 4 bisecants : { 3, 5, 21, 23 }
Off point 27 = $P_{37} = (0, 3, 1)$ lies on 4 bisecants : { 6, 8, 12, 17 }
Off point 28 = $P_{39} = (2, 3, 1)$ lies on 4 bisecants : { 13, 14, 23, 25 }
Off point 29 = $P_{40} = (3, 3, 1)$ lies on 4 bisecants : { 9, 11, 18, 20 }
Off point 30 = $P_{41} = (4, 3, 1)$ lies on 4 bisecants : { 2, 5, 22, 26 }
Off point 31 = $P_{44} = (7, 3, 1)$ lies on 4 bisecants : { 0, 3, 15, 24 }
Off point 32 = $P_{45} = (8, 3, 1)$ lies on 4 bisecants : { 4, 7, 16, 21 }
Off point 33 = $P_{46} = (0, 4, 1)$ lies on 4 bisecants : { 2, 11, 12, 24 }
Off point 34 = $P_{48} = (2, 4, 1)$ lies on 4 bisecants : { 0, 4, 6, 10 }
Off point 35 = $P_{49} = (3, 4, 1)$ lies on 4 bisecants : { 5, 8, 14, 19 }
Off point 36 = $P_{50} = (4, 4, 1)$ lies on 4 bisecants : { 3, 7, 13, 26 }
Off point 37 = $P_{53} = (7, 4, 1)$ lies on 4 bisecants : { 1, 9, 16, 22 }
Off point 38 = $P_{54} = (8, 4, 1)$ lies on 4 bisecants : { 17, 20, 21, 25 }
Off point 39 = $P_{55} = (0, 5, 1)$ lies on 4 bisecants : { 7, 10, 12, 15 }
Off point 40 = $P_{56} = (1, 5, 1)$ lies on 4 bisecants : { 3, 6, 20, 22 }
Off point 41 = $P_{59} = (4, 5, 1)$ lies on 4 bisecants : { 17, 19, 23, 26 }
Off point 42 = $P_{60} = (5, 5, 1)$ lies on 4 bisecants : { 4, 5, 24, 25 }
Off point 43 = $P_{61} = (6, 5, 1)$ lies on 4 bisecants : { 8, 11, 13, 16 }
Off point 44 = $P_{63} = (8, 5, 1)$ lies on 4 bisecants : { 1, 2, 18, 21 }
Off point 45 = $P_{64} = (0, 6, 1)$ lies on 4 bisecants : { 1, 3, 12, 14 }
Off point 46 = $P_{66} = (2, 6, 1)$ lies on 4 bisecants : { 17, 18, 22, 24 }
Off point 47 = $P_{67} = (3, 6, 1)$ lies on 4 bisecants : { 2, 4, 13, 15 }
Off point 48 = $P_{68} = (4, 6, 1)$ lies on 4 bisecants : { 9, 10, 25, 26 }
Off point 49 = $P_{71} = (7, 6, 1)$ lies on 4 bisecants : { 7, 8, 20, 23 }
Off point 50 = $P_{72} = (8, 6, 1)$ lies on 4 bisecants : { 0, 11, 19, 21 }
Off point 51 = $P_{73} = (0, 7, 1)$ lies on 4 bisecants : { 0, 5, 12, 20 }
Off point 52 = $P_{74} = (1, 7, 1)$ lies on 4 bisecants : { 1, 8, 15, 25 }
Off point 53 = $P_{77} = (4, 7, 1)$ lies on 4 bisecants : { 14, 16, 24, 26 }
Off point 54 = $P_{78} = (5, 7, 1)$ lies on 4 bisecants : { 10, 11, 22, 23 }
Off point 55 = $P_{79} = (6, 7, 1)$ lies on 4 bisecants : { 3, 4, 18, 19 }
Off point 56 = $P_{81} = (8, 7, 1)$ lies on 4 bisecants : { 6, 9, 13, 21 }
Off point 57 = $P_{82} = (0, 8, 1)$ lies on 4 bisecants : { 4, 9, 12, 23 }
Off point 58 = $P_{84} = (2, 8, 1)$ lies on 4 bisecants : { 1, 5, 7, 11 }
Off point 59 = $P_{85} = (3, 8, 1)$ lies on 4 bisecants : { 3, 10, 16, 17 }
Off point 60 = $P_{86} = (4, 8, 1)$ lies on 4 bisecants : { 0, 8, 18, 26 }
Off point 61 = $P_{89} = (7, 8, 1)$ lies on 4 bisecants : { 2, 6, 19, 25 }
Off point 62 = $P_{90} = (8, 8, 1)$ lies on 4 bisecants : { 14, 15, 21, 22 }

The gradient

The gradient of the quartic curve is :

$$8X_2^3$$

$$(0,0,8,0,0,0,0,0,0,0)$$

$$2X_1^3$$

$$(0,2,0,0,0,0,0,0,0,0)$$

$$3X_0^3$$

$$(3,0,0,0,0,0,0,0,0,0)$$